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GREEN PEACE LINCOLN COLLEGE

(Affiliated to Lincoln University College, Malaysia)

Itahari, Sunsari, Nepal

FACULTY OF MBA/BBA/BIT

ASSIGNMENT COVER PAGE

Please fill in all the required details for your assignment to be accepted.

Student's Name	Bishnu	Sah		
Year/Semester	Third	Sixth	**Intake**	2023
Program	BIT			
Subject Name/ Code	Mobile Computing	BIT 351		
LC Number	LC00023000806			
Lecturer's Name	Yogendra raj	Shrestha		
Assignment Tittle (1) ☐ (2) ✓	Lab Report			
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DECLARATION BY STUDENTS:

*I Certify that assignment is my own work in my own words. All resources have been acknowledged and the content has not been previously submitted for assessment to **LINCOLN** or elsewhere. I also confirm that I have kept a copy of this assignment.*

Signed:

Date: 0 3 0 9 2 0 2 6

Lab 1 Title: IP address configuration (static & dynamic) 1.Objective

- To ensure no two devices on the same network share an address (avoiding conflicts).
- To allow data packets to reach the correct destination across local and wide-area networks.
- To enable devices to communicate with each other and with external networks/the internet.

2.Theory

IP address configuration ensures every device on a network has a unique address to communicate properly. Static IP configuration means manually assigning a fixed address that never changes used for servers, printers, and routers that need to stay easily reachable. Dynamic IP configuration uses DHCP to automatically assign addresses from a shared pool on a temporary lease basis used for devices like laptops and phones that frequently connect and disconnect.

3.Tools

- Cisco Packet Tracer
- 2 PCs / Laptops
- 1 Switch
- Copper straight

4.Procedure

I. Put two PCs and one switch in Cisco Packet Tracer. Connect the PCs to the switch using cables.

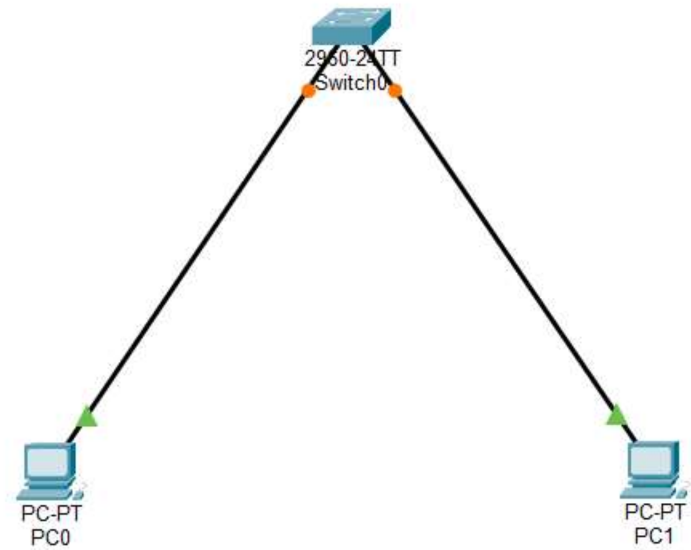
II. Open PC0 → Desktop → IP Configuration. Select Static and enter IP 192.168.1.1 and subnet mask 255.255.255.0.

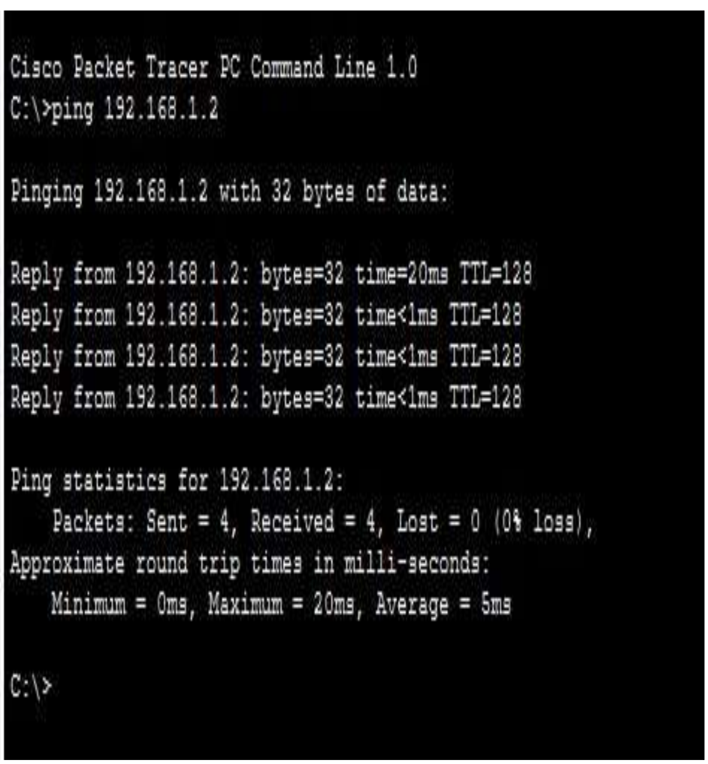
III. Open PC1 → IP Configuration. Select Static and enter IP 192.168.1.2 and subnet mask 255.255.255.0.

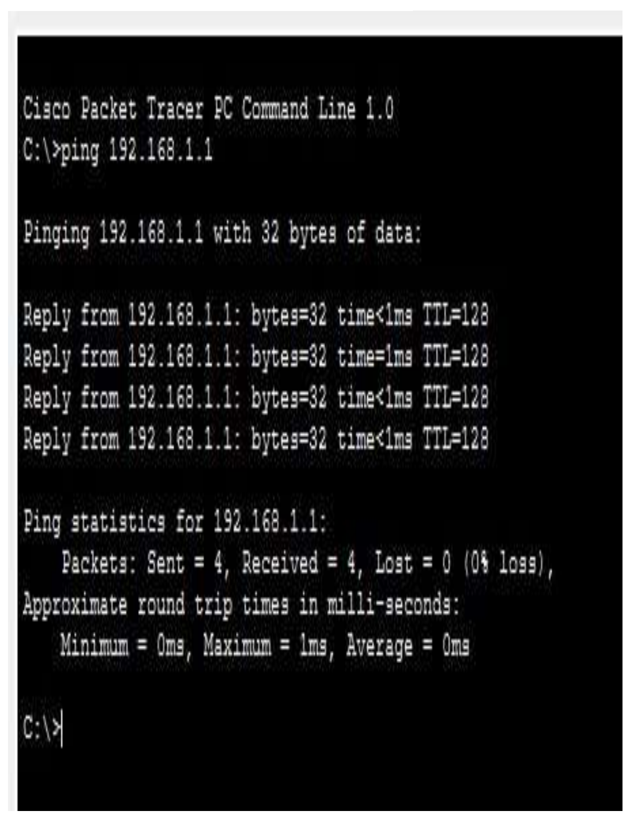
IV. Open the Command Prompt on PC0 and use ping 192.168.1.20 to check the connection.

V. Change PC1 from Static to DHCP and see how it gets an IP address automatically.

5.Screenshots







6. Results

Both PCs were given static IP addresses on the same network. The connection between the PCs was tested successfully using the ping command. PC1 was also changed to DHCP and received an IP address automatically. This showed the difference between static and dynamic IP addressing.

7. Conclusion both static and dynamic IP configuration serve the fundamental purpose of giving devices unique addresses so they can communicate effectively on a network, but they achieve this through different approaches suited to different needs. Static configuration offers stability and predictability, making it ideal for servers, printers, and routers that must remain consistently reachable, while dynamic configuration through DHCP offers automation and efficient address management, making it ideal for end-user devices that frequently join and leave the network.

Lab 2 Title: : DHCP Server Configuration and Testing 1. Objective

- To configure the server to automatically assign IP addresses, subnet masks, default gateways, and DNS setting.
- To test that client devices correctly receive valid IP configurations from the DHCP server without manual input.
- To confirm that DHCP-assigned devices can communicate within the network and access external resources (e.g., the internet) without conflicts or errors.

2. Theory

DHCP (Dynamic Host Configuration Protocol) is a client-server protocol that automatically assigns IP addresses and other network configuration settings (subnet mask, default gateway, DNS) to devices on a network, eliminating the need for manual configuration.

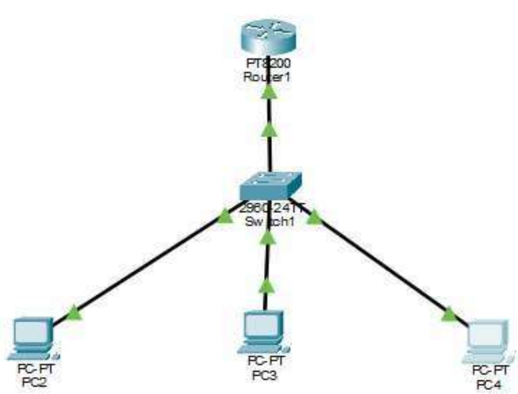
3. Tools

- Cisco Packet Tracer
- • 1 Router
- • 1 Switch
- • 2–3 PCs
- • Connecting cables

4. Procedure

I. Add a router, switch, and two or three PCs to the Packet Tracer workspace. Connect them using cables.
II. Configure the router as a DHCP server and set the network details.
III. Open the IP Configuration of each PC and select DHCP.
IV. Check that each PC gets an IP address automatically.
V. Use the ping command to check whether the PCs can communicate with each other.

5. Screenshots



```
Router>en
Router#con t
Router#con ter
Router#config ter
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#inter
Router(config)#interface fast
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ex
Router(config-if)#exit
Router(config)#ip dhcp pool
Router(config)#ip dhcp pool 1
Router(config)#ip dhcp pool LAB_POOL
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#defa
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#ex
Router(config)#ip dhcp excluded
Router(config)#ip dhcp excluded-address
Router(config)#ip dhcp excluded-address 192.168.1.1
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

6. Result

The router was successfully configured as a DHCP server. All PCs automatically received valid IP addresses and network settings. Connectivity was verified successfully using the ping command. The DHCP assignments were checked and found to be working correctly. All connected devices were able to communicate successfully.

7. Conclusion

Configuring and testing a DHCP server confirms that IP addresses and network settings can be assigned to client devices automatically and reliably, without manual intervention. This not only saves time and reduces configuration errors but also ensures smooth communication between devices and proper connectivity to network resources, demonstrating the practical value of DHCP in managing modern, scalable networks.

Lab 3

Title: Router Configuration (Basic Routing)

1. Objective

- To configure a router to forward data packets between two or more separate networks
- To set up routing tables (static or dynamic) so data reaches its correct destination efficiently.
- To verify that devices on different networks connected via the router can successfully communicate with each other.

2 .Theory

A router connects multiple networks and forwards data packets between them based on destination IP addresses, operating at Layer 3 (Network Layer) of the OSI model. Each router interface is assigned an IP address belonging to a different network, and the router uses a routing table to decide where to send incoming packets. Routes can be added statically (manually entered by an administrator) or dynamically (automatically learned using routing protocols like RIP or OSPF). When a packet arrives, the router matches its destination address to an entry in the routing table and forwards it out the correct interface, hop by hop, until it reaches its destination network.

3. Tools

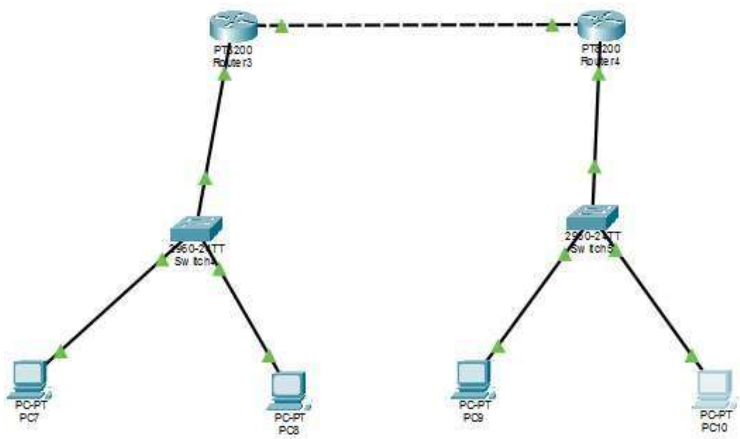
- Cisco Router – 2
- Switch – 2
- PC – 4
- Copper Straight
- Cisco Packet Tracer

4. Procedure

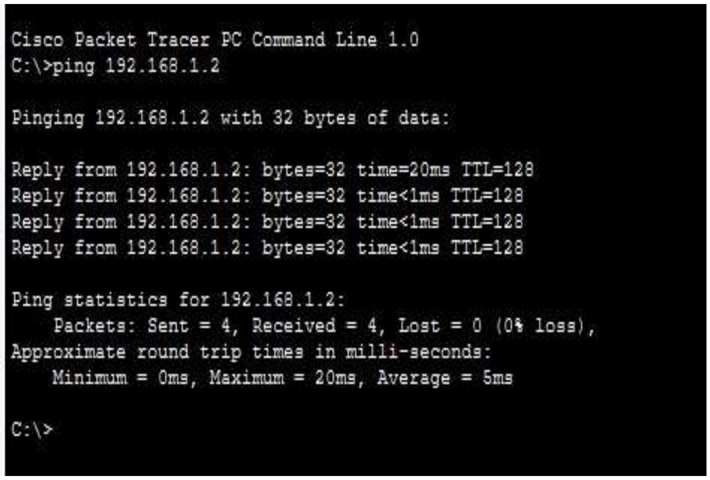
I. Place two routers, two switches, and four PCs in the workspace.
II. Connect the device and configure all the pc.
III. Configure both router 1 and 2
IV. Add static routes on both routers

V. Check the routing tables and test connectivity between pc on different network. **5.**

Screenshots




```
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip add
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface f
Router(config-if)#interface fast
Router(config-if)#interface fast
Router(config-if)#ex
Router(config)#interface f
Router(config)#interface fastEthernet 0/1
Router(config-if)#io address 192.168.2.1 255.255.255.0
                   ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#ex
Router(config)#do show ip interface br
Router(config)#do show ip interface brief
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        192.168.2.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=20ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>
```

6. Results

The router's two interfaces were configured on separate networks, and basic routing between them worked as confirmed by ping.

7. Conclusion

Basic router configuration shows how routers enable communication between separate networks by using routing tables to direct data toward its correct destination. Whether achieved through static routes for simple, controlled setups or dynamic routing protocols for larger, changing networks, proper configuration ensures accurate and efficient packet delivery, making the router a fundamental building block of any multi-network infrastructure.

Lab 4

Title: Configuration of a Wi-Fi Access Point (AP)

1.Objective

- To configure the AP to allow devices to connect to the network wirelessly instead of using physical cables.
- To set up authentication methods (e.g., WPA2/WPA3, passwords) to prevent unauthorized users from accessing the network.
- To configure SSID, channel, and IP settings so the AP works correctly with the existing wired network and other devices.

2. Theory

Configuring a Wi-Fi Access Point involves setting up a device that bridges wireless clients to a wired network by broadcasting a unique network name (SSID), operating on specific wireless standards and frequency bands (2.4 GHz or 5 GHz) with an appropriate channel to minimize interference, and securing the connection using encryption protocols like WPA2/WPA3 with a password to prevent unauthorized access. The AP typically integrates with the existing network's DHCP server so connected devices automatically receive IP addresses, and it must be properly positioned and configured for transmission power and channel management to ensure stable, reliable wireless coverage across the intended area together enabling seamless, secure, and efficient wireless connectivity.

3. Tools

- Cisco Packet Tracer
- 1 Wireless Access Point
- 2 Wireless-enabled laptops or PCs
- Router or switch

4. Procedure

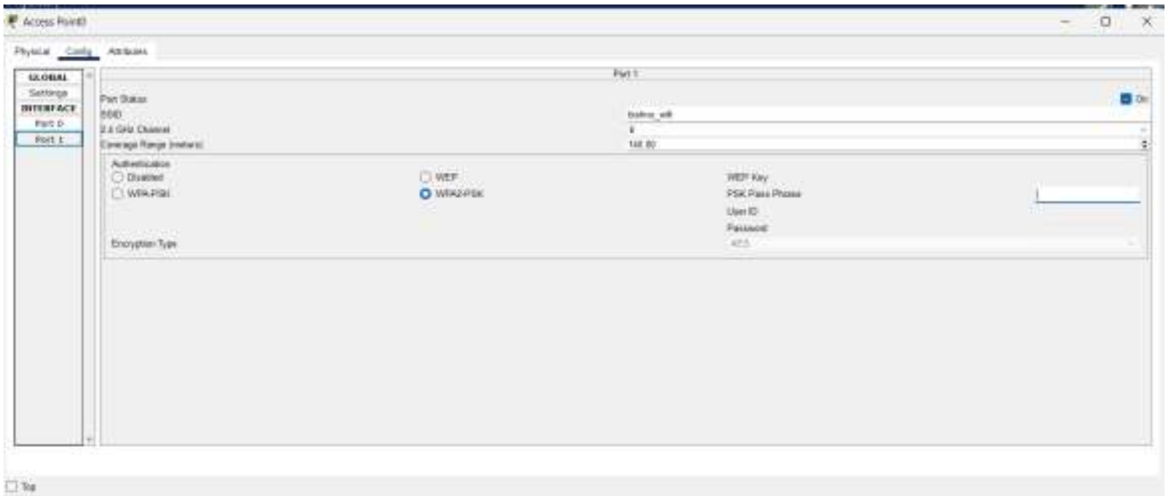
I. Add a Wireless Access Point and two laptops to the Cisco Packet Tracer workspace. Install wireless modules in the laptops if required.

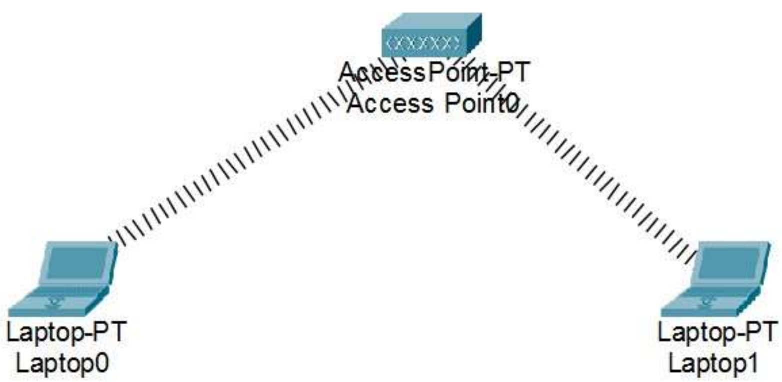
II. Open the AP configuration and set an SSID such as Lab_WiFi.

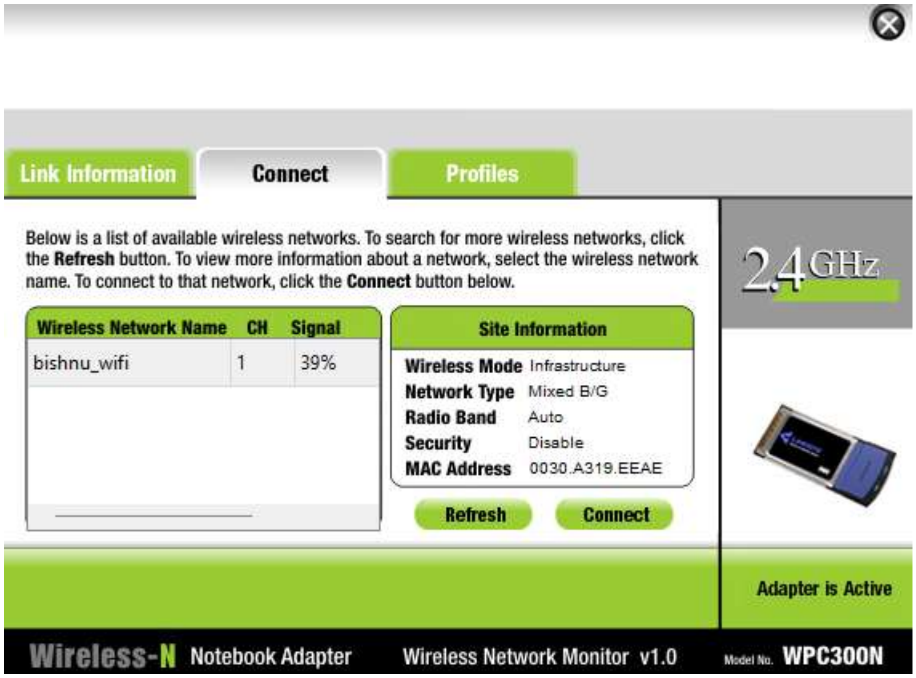
III. Open Desktop PC Wireless on each laptop and connect them to the configured Wi-Fi network.

IV. Configure suitable IP addresses for both laptops within the same network.

V. Use the ping command between the laptops to check whether they can communicate successfully.







6. Results

The results confirmed that the Wi-Fi Access Point was configured successfully, broadcasting a unique SSID that nearby devices could detect and connect to, while WPA2/WPA3 security effectively blocked unauthorized access attempts by requiring the correct password. Client devices that connected to the AP automatically received valid IP addresses through the network's DHCP service, and were able to communicate with other devices as well as access external resources like the internet, confirming smooth integration between the wireless and wired network.

7. Conclusion

The configuration and testing of the Wi-Fi Access Point confirmed that it successfully provides secure, reliable wireless connectivity while properly integrating with the existing wired network. The AP's SSID broadcast, WPA2/WPA3 security, and automatic IP assignment worked together to allow authorized devices to connect seamlessly and access network resources, demonstrating that the AP configuration met its intended objectives of enabling secure, stable, and efficient wireless communication.

Lab 5

Title: Wireless Network Security (WPA2/WPA3, MAC Filtering)

1.Objective

- To implement encryption protocols (WPA2/WPA3) to protect data transmitted over the wireless network from interception or eavesdropping.
- To configure the access point to allow or deny connections based on specific device MAC addresses, adding an extra layer of access control.
- To ensure that the combination of encryption and MAC filtering minimizes vulnerabilities and protects the network from common wireless security threats.

2. Theory

Wireless network security relies on encryption and access control to protect data transmitted over open radio frequencies. WPA2 uses AES encryption with a pre-shared key (password) and a 4-way handshake to secure connections, while WPA3 improves on this with SAE (Simultaneous Authentication of Equals), offering stronger protection against offline password-guessing attacks and ensuring forward secrecy. MAC filtering adds an extra layer by allowing only devices with approved MAC addresses to connect, though it can be bypassed through MAC spoofing.

3. Tools

- Cisco Packet Tracer
- 1 Wireless Access Point
- 2 Laptops/PCs with wireless adapters

4. Procedure

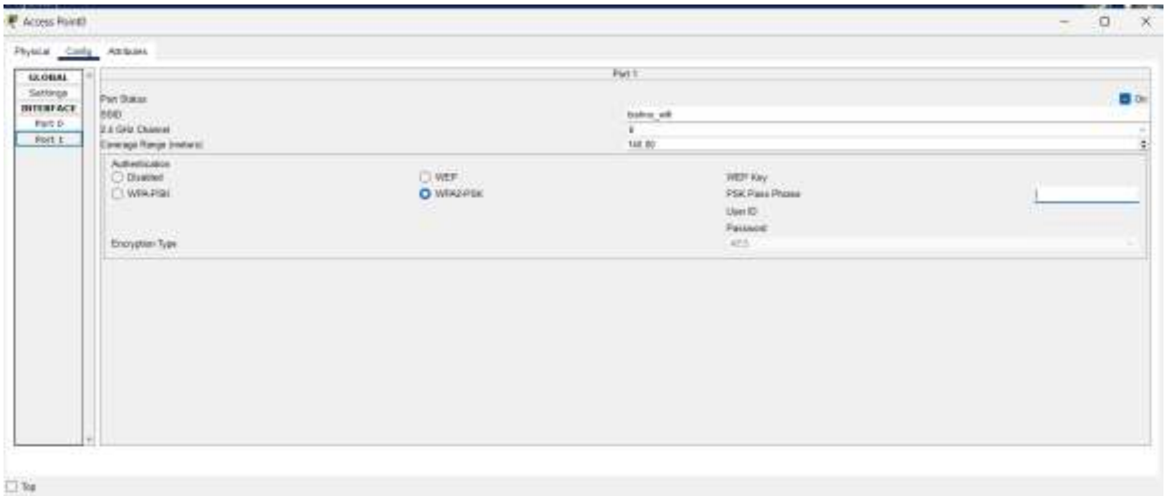
I. Open the Access Point used in the previous practical and configure wireless security using WPA2-PSK or WPA3-Personal. Set a suitable password, such as PrajwalWifi@999

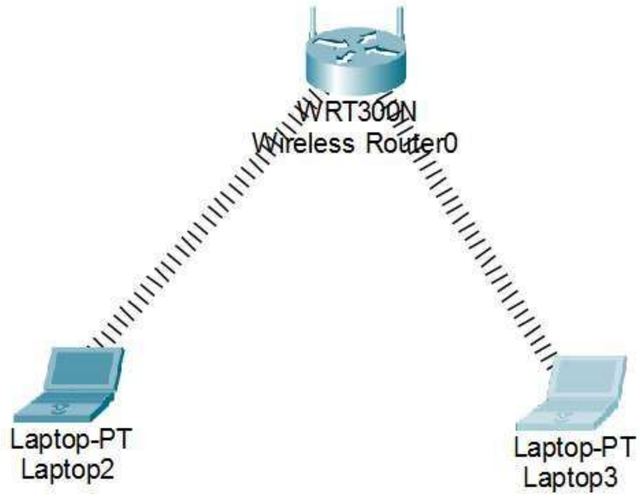
II. Connect both laptops to the configured SSID using the assigned password.

III. Open the MAC filtering settings on the Access Point and enter the MAC addresses of the authorized laptops.

IV. Test the connection by confirming that authorized devices can access the network while an unregistered device is denied access.

5.Screenshots






```
C:\>
C:\>
C:\>ping 192.168.0.102

Pinging 192.168.0.102 with 32 bytes of data:

Reply from 192.168.0.102: bytes=32 time=79ms TTL=128
Reply from 192.168.0.102: bytes=32 time=48ms TTL=128
Reply from 192.168.0.102: bytes=32 time=40ms TTL=128
Reply from 192.168.0.102: bytes=32 time=36ms TTL=128

Ping statistics for 192.168.0.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 36ms, Maximum = 79ms, Average = 50ms

C:\>
```

6. Results

WPA2/WPA3 encryption was successfully configured, and devices without the correct password were denied access, confirming secure authentication. MAC filtering was tested by allowing only approved devices to connect, while devices with unregistered MAC addresses were successfully blocked, confirming the additional access control layer worked as intended.

7. Conclusion

The combination of WPA2/WPA3 encryption and MAC filtering successfully secured the wireless network by protecting data transmission and restricting access to authorized users and devices. This layered security approach effectively met the objective of preventing unauthorized access and enhancing overall wireless network integrity.